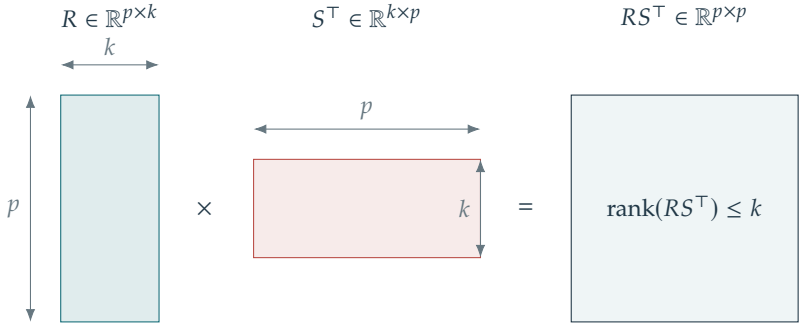
$R \in \mathbb{R}^{p \times k}$
$S^\top \in \mathbb{R}^{k \times p}$
$RS^\top \in \mathbb{R}^{p \times p}$
k
p
p
k
$\times$
=
$\text{rank}(RS^\top) \leq k$
$x \in \mathbb{R}^p$
compress
$S^\top x \in \mathbb{R}^k$
reconstruct
$RS^\top x \in \mathbb{R}^p$